SET light propagation

Clocks delay and deflection

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The observer's clock participates in the field it is used to measure. In SET, its ticks are physical events whose progression depends on the available field interaction capacity. The same burden that enters the clock relation also enters signal propagation. We follow that connection from one spherical source to clock comparisons, the travel time and deflection of light, and the frequency observable measured in the Cassini solar-conjunction experiment.

We retain the stationary spherical response and probe coupling developed in SET perihelion derivation 17 [1]. The calculations below use that response throughout. They give the standard spherical light-propagation predictions, including a Cassini comparison through the published parameter estimate. Their purpose is to show how several measurements follow from the same SET construction without a separate adjustment for each observable.

1 The source and the retained exterior

Start with the capacity relation. Here S is gravitational burden expressed in speed units, V is physical displacement per stationary reference time T, and $\beta = d\tau/dT$ is the probe's event progression. A held clock has $V = 0$ and rate α:

$$S^2 + V^2 + c^2\beta^2 = c^2, \quad \alpha^2 = 1 - \frac{S^2}{c^2}. \quad (1)$$

For a spherical source domain of radius R_0, SET's source law gives

$$V_\Omega = \frac{4\pi R_0^3}{3}, \quad Q_\Omega^2 = 24\pi GMV_\Omega = 32\pi^2 GMR_0^3, \quad (2)$$

$$\mu_Q \equiv \frac{Q_\Omega^2}{24\pi V_\Omega} = \frac{Q_\Omega^2}{32\pi^2 R_0^3} = GM. \quad (3)$$

The source radius R_0 and measurement radius r remain distinct. In the isolated exterior retained in [1], with T normalized to an unloaded clock at infinity,

$$S^2(r) = \frac{2\mu_Q}{r}, \quad A(r) \equiv \alpha^2(r) = 1 - \frac{r_s}{r}, \quad r_s = \frac{2\mu_Q}{c^2}. \quad (4)$$

The corresponding spatial response is

$$d\ell^2 = \frac{dr^2}{A} + r^2 d\Omega^2, \quad d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2. \quad (5)$$

Equations (4) and (5) specify the spherical field used here. Derivation 17 obtains their correspondence within its stated source-boundary and conserved work-current construction. These are the shared source and propagation relations used for every observable below. All held-clock expressions below apply where $A > 0$.

For a physical trajectory, $V^2 = d\ell^2/dT^2$. Multiplying equation (1) by dT^2 therefore gives

$$c^2 d\tau^2 = c^2 A\, dT^2 - \frac{dr^2}{A} - r^2 d\Omega^2 . \quad (6)$$

The interval records the operational clock and distance relations. SET interprets them through participation in field capacity. Using this representation does not require us to replace that interpretation with a different physical picture.

2 What clocks and local observers measure

On the light branch, $d\tau = 0$. Equations (1) and (5) give

$$\frac{d\ell}{dT} = c\alpha, \quad \frac{d\tau_{\text{local}}}{dT} = \alpha, \quad \frac{d\ell}{d\tau_{\text{local}}} = c . \quad (7)$$

Light covers less local distance per reference second where the held-clock factor is smaller. The local clock records fewer ticks per reference second by that same factor. Their ratio remains c. The common factor cancels in a local measurement; comparisons between locations and travel times referred to a specified clock retain the gravitational effect.

For radial light, the additional distance relation $d\ell = |dr|/\alpha$ gives

$$\left|\frac{dr}{dT}\right| = c\alpha^2 = cA, \quad n_r \equiv \frac{c}{|dr/dT|} = \frac{1}{A} . \quad (8)$$

The radial coordinate index n_r contains both the clock and spatial relations. Keeping that distinction prevents a factor from being counted twice or omitted. With the held-clock-normalized burden $q = S/\alpha$, the same index is

$$n_r = 1 + \frac{q^2}{c^2}, \quad q = \frac{S}{\alpha}, \quad Q_\star(r) = 4\pi r^2 q . \quad (9)$$

$Q_\star$ is a clock-normalized screen reading, not a new source strength. The fixed source quantity is Q_Ω at R_0. In particular, the screen reading $4\pi r^2 S(r)$ varies with r even though μ_Q is fixed.

A static frequency comparison follows directly. Let a held emitter produce successive crests a proper interval $\Delta\tau_e$ apart. Their emission interval in T is $\Delta T = \Delta\tau_e/\alpha_e$. In a stationary field between fixed endpoints, both crests acquire the same travel time, so their arrival separation in T is unchanged. A held receiver records $\Delta\tau_r = \alpha_r \Delta T$. Hence

$$\frac{\nu_r}{\nu_e} = \frac{\alpha_e}{\alpha_r} = \sqrt{\frac{1 - S_e^2/c^2}{1 - S_r^2/c^2}} . \quad (10)$$

A photon sent outward from a more burdened location is therefore received at a lower local frequency. A fixed propagation delay does not create an additional frequency drift. A changing path delay does, as we calculate below.

3 The optical path from the probe coupling

To determine a light path as well as its local speed, use the retained probe action $L = -mc^2\beta$. At a point, write p for the magnitude of momentum in the physical spatial reading. The capacity relation gives

$$p = \frac{mV}{\beta}, \quad E = pV - L = \frac{mc^2\alpha^2}{\beta}, \quad E^2 = \alpha^2(m^2c^4 + c^2p^2). \quad (11)$$

Take the limit $m \to 0$ while keeping nonzero momentum finite. The resulting Hamiltonian is $E = c\alpha p$. In spatial coordinates, $p^2 = q^{ij}p_ip_j$, where q_{ij} is the distance relation in equation (5). Thus the null Hamiltonian and stationary abbreviated action are

$$\mathscr{H} = c\alpha\sqrt{q^{ij}p_ip_j}, \quad I_{\mathsf{opt}} = \int p_i\,dx^i = \frac{E}{c}\int_\Gamma \frac{d\ell}{\alpha}. \quad (12)$$

This limit preserves the equations of motion; setting $m = 0$ in the massive Lagrangian before varying it would instead erase the action. At fixed reference energy and fixed endpoints, stationarity of equation (12) gives the travel-time functional

$$\mathscr{T}[\Gamma] = \frac{1}{c}\int_\Gamma \frac{d\ell}{\alpha}, \quad \delta\mathscr{T} = 0. \quad (13)$$

For an equatorial path written as $r(\varphi)$, its exact form is

$$\mathscr{T} = \frac{1}{c}\int\sqrt{\frac{1}{A^2}\left(\frac{dr}{d\varphi}\right)^2 + \frac{r^2}{A}}\,d\varphi. \quad (14)$$

The local cancellation in equation (7) and the accumulated delay in equation (13) describe different measurements of the same field. The first uses one local clock; the second compares events connected by a path through varying burden. Ray deflection follows by varying that path.

As a direct check, radial propagation between fixed exterior radii $r_1 < r_2$ integrates to

$$\mathscr{T}_{12} = \frac{r_2 - r_1}{c} + \frac{r_s}{c}\ln\left(\frac{r_2 - r_s}{r_1 - r_s}\right). \quad (15)$$

For an immediate reflection along this same radial path, the reference round-trip time is $2\mathscr{T}_{12}$, and the held clock at r_1 records $2\alpha(r_1)\mathscr{T}_{12}$. A solar conjunction requires a nonradial path, which we now calculate.

4 Delay along a nonradial path

To compare the bent path with a straight reference path, introduce isotropic radius ρ. It is a coordinate change within the retained exterior, defined by

$$r = \rho(1 + h)^2, \quad h = \frac{\mu_Q}{2c^2\rho}, \quad \alpha = \frac{1 - h}{1 + h}. \quad (16)$$

Substitution into equation (5) gives $d\ell = (1+h)^2 ds$, where $ds^2 = d\rho^2 + \rho^2 d\Omega^2$ is the Euclidean coordinate line element. The optical index referred to ds is therefore

$$\mathscr{T} = \frac{1}{c}\int_\Gamma n_{\mathsf{opt}}\, ds, \quad n_{\mathsf{opt}} = \frac{(1+h)^3}{1-h} = 1 + \frac{2\mu_Q}{c^2\rho} + O\!\left(\frac{\mu_Q^2}{c^4\rho^2}\right). \quad (17)$$

n_{opt} and the areal radial index n_r use different coordinate lengths. Both come from the same $d\ell/\alpha$. The first-order coefficient in equation (17) includes the clock response and the spatial response; using $1/\alpha$ with an unchanged Euclidean distance would omit the latter.

Work to first order in the source strength for fixed endpoints. The zeroth-order straight path is already stationary in flat space, so its first-order displacement does not change the zeroth-order travel time. We may evaluate the first-order index correction along that straight path. Let b be its impact parameter, let z run along it, and place the endpoints on opposite sides of the source. Then $\rho(z) = \sqrt{b^2 + z^2}$ and

$$L_E = \sqrt{\rho_E^2 - b^2}, \quad L_P = \sqrt{\rho_P^2 - b^2}\,. \quad (18)$$

$$\Delta\mathscr{T}_{\mathsf{one}} = \frac{2\mu_Q}{c^3}\int_{-L_E}^{L_P} \frac{dz}{\sqrt{b^2+z^2}} = \frac{2\mu_Q}{c^3}\ln\left[\frac{(\rho_E + L_E)(\rho_P + L_P)}{b^2}\right]. \quad (19)$$

The excess is defined relative to the straight vacuum coordinate time $(L_E + L_P)/c$. For fixed endpoints and a reciprocal round trip, it is twice equation (19). When b is small compared with both endpoint radii,

$$\Delta\mathscr{T}_{\mathsf{two}} \simeq \frac{4\mu_Q}{c^3}\ln\left(\frac{4\rho_E\rho_P}{b^2}\right) = \frac{Q_\Omega^2}{8\pi^2 R_0^3 c^3}\ln\left(\frac{4\rho_E\rho_P}{b^2}\right). \quad (20)$$

This is the leading Shapiro delay in source-law variables. Q_Ω fixes its coefficient through equation (3); the chosen path fixes the logarithm. The source radius has not become an impact parameter, and no additional optical source strength has been fitted.

5 A changing delay becomes a frequency shift

Let T_e and T_r denote emission and reception of a phase crest in a coherent link, after accounting for the known transponder frequency ratio. The travel time $\mathscr{T}$ may depend on the emission epoch through the endpoint geometry. Phase conservation gives the reference-time frequency relation

$$T_r = T_e + \mathscr{T}(T_e), \quad \frac{\nu_r^{(T)}}{\nu_e^{(T)}} = \frac{dT_e}{dT_r} = \left(1 + \frac{d\mathscr{T}}{dT_e}\right)^{-1}. \quad (21)$$

Here the superscript (T) identifies frequencies per reference second. Separate $\mathscr{T}$ into its reference propagation time and gravitational excess. Expanding the reciprocal and removing the ordinary kinematic Doppler contribution gives, to leading order,

$$y_{\mathsf{grav}} \equiv \frac{\nu_{\mathsf{received}} - \nu_{\mathsf{reference}}}{\nu_{\mathsf{reference}}} \simeq -\frac{d\Delta\mathscr{T}_{\mathsf{two}}}{dT}. \quad (22)$$

We define positive y as a received frequency increase. An increasing delay separates arriving crests and gives negative y. For a held Earth station with constant clock factor, converting both emission and reception times to its clock leaves this fractional relation unchanged. The actual moving link also requires endpoint motion, clock conversions and uplink/downlink geometry.

Differentiating equation (20), before holding the endpoint radii fixed, gives

$$y_{\mathrm{grav}} \simeq \frac{8\mu_Q}{c^3}\frac{\dot{b}}{b} - \frac{4\mu_Q}{c^3}\left(\frac{\dot{\rho}_E}{\rho_E} + \frac{\dot{\rho}_P}{\rho_P}\right). \quad (23)$$

Near conjunction the rapidly changing impact parameter usually supplies the dominant term. For fixed endpoint radii the finite-endpoint version from equation (19) is

$$y_{\mathrm{grav}} = \frac{4\mu_Q}{c^3}\frac{\dot{b}}{b}\left(\frac{\rho_E}{L_E} + \frac{\rho_P}{L_P}\right) \simeq \frac{8\mu_Q}{c^3}\frac{\dot{b}}{b}. \quad (24)$$

The final approximation becomes $y_{\mathrm{grav}} \simeq Q_\Omega^2 \dot{b}/(4\pi^2 R_0^3 c^3 b)$ in the source variables. The derivative, not the total accumulated delay, is the frequency observable.

For an illustrative symmetric conjunction, choose $b(T) = \sqrt{b_0^2 + v_\perp^2 T^2}$. The large-endpoint expression becomes

$$y_{\mathrm{grav}}(T) \simeq \frac{8\mu_Q}{c^3}\frac{v_\perp^2 T}{b_0^2 + v_\perp^2 T^2}. \quad (25)$$

At closest approach the delay is largest, but its derivative is zero. The frequency contribution changes sign across conjunction. This example makes the measurement distinction visible without interpreting a constant delay as a continuing loss of photon frequency.

6 Deflection from the same optical response

Varying $\int n_{\mathrm{opt}} ds$ gives the ray equation $d(n_{\mathrm{opt}}\boldsymbol{t})/ds = \nabla n_{\mathrm{opt}}$, where $\boldsymbol{t}$ is the unit coordinate tangent. For a weakly deflected ray initially along z, integrate the transverse gradient along the straight reference path. At positive impact parameter b,

$$\Delta t_\perp \simeq \int_{-\infty}^{\infty} \frac{\partial n_{\mathrm{opt}}}{\partial b}\, dz = -\frac{2\mu_Q b}{c^2}\int_{-\infty}^{\infty} \frac{dz}{(b^2+z^2)^{3/2}} = -\frac{4\mu_Q}{c^2 b}. \quad (26)$$

The negative sign points toward the source. The bending angle for distant endpoints is therefore

$$\delta\theta = \frac{4\mu_Q}{c^2 b} = \frac{Q_\Omega^2}{8\pi^2 R_0^3 c^2 b}. \quad (27)$$

The same result follows from the exact null Hamiltonian. Define $b_\infty = cL/E$ using the conserved angular momentum L, set $w = 1/r$, and use a prime for $d/d\varphi$. Eliminating the radial momentum gives

$$(w')^2 = \frac{1}{b_\infty^2} - w^2 + \frac{2\mu_Q}{c^2}w^3, \quad w'' + w = \frac{3\mu_Q}{c^2}w^2. \quad (28)$$

The first-order solution has the deflection in equation (27). At this order the straight-path b and asymptotic b_∞ can be identified. This provides an independent calculation through the same orbit machinery used for massive probes, rather than a second fitted bending rule.

7 Numerical calculation and the Cassini comparison

For a reproducible solar benchmark, use the IAU nominal values $\mu_\odot = 1.3271244 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$ and $R_\odot = 6.957 \times 10^{8}\,\mathrm{m}$, with $c = 299792458\,\mathrm{m\,s^{-1}}$ and $1\,\mathrm{au} = 149597870700\,\mathrm{m}$ [3]. Nominal solar values are conversion constants close to measured solar properties; they are not uncertainty-free measurements of the Sun.

$$Q_{\Omega,\odot} = 4\pi\sqrt{2\mu_\odot R_\odot^3} = 3.7567620 \times 10^{24}\,\mathrm{m^3\,s^{-1}}, \quad r_s = 2953.2501\,\mathrm{m}. \quad (29)$$

The source value above uses $R_0 = R_\odot$. For held clocks at the nominal solar surface and at one au, the predicted frequency ratio is $\alpha_\odot/\alpha_{\mathrm{au}} = 0.9999978874$. The leading grazing deflection is 1.751190 arcseconds. These are calculated benchmarks; we do not label them new measurements.

Table 1 uses fixed isotropic endpoint radii $\rho_E = 1\,\mathrm{au}$ and $\rho_P = 8\,\mathrm{au}$. The delay is the finite-endpoint first-order round-trip result, and the bending column uses distant endpoints. The geometry is illustrative and is not an ephemeris reconstruction of Cassini.

Impact parameter	Round-trip excess delay	Leading distant-source deflection
1.6 solar radii	261.392 microseconds	1.094494 arcseconds
3 solar radii	236.621 microseconds	0.583730 arcseconds
6 solar radii	209.306 microseconds	0.291865 arcseconds

Table 1. Calculated values at fixed nominal solar source strength. The number of digits supports reproducibility and does not assign observational precision to the illustration.

For Figure 1, use the same endpoint radii with $b_0 = 1.6R_\odot$ and $v_\perp = 30\,\mathrm{km\,s^{-1}}$. The finite-endpoint expressions produce a maximum delay of 261.392 microseconds. The leading frequency extrema occur near $T = \pm\, b_0/v_\perp = \pm\, 0.42944$ days, with magnitude about 5.31×10^{-10}. The plots are predictions for this stated geometry, with no observational points attached.

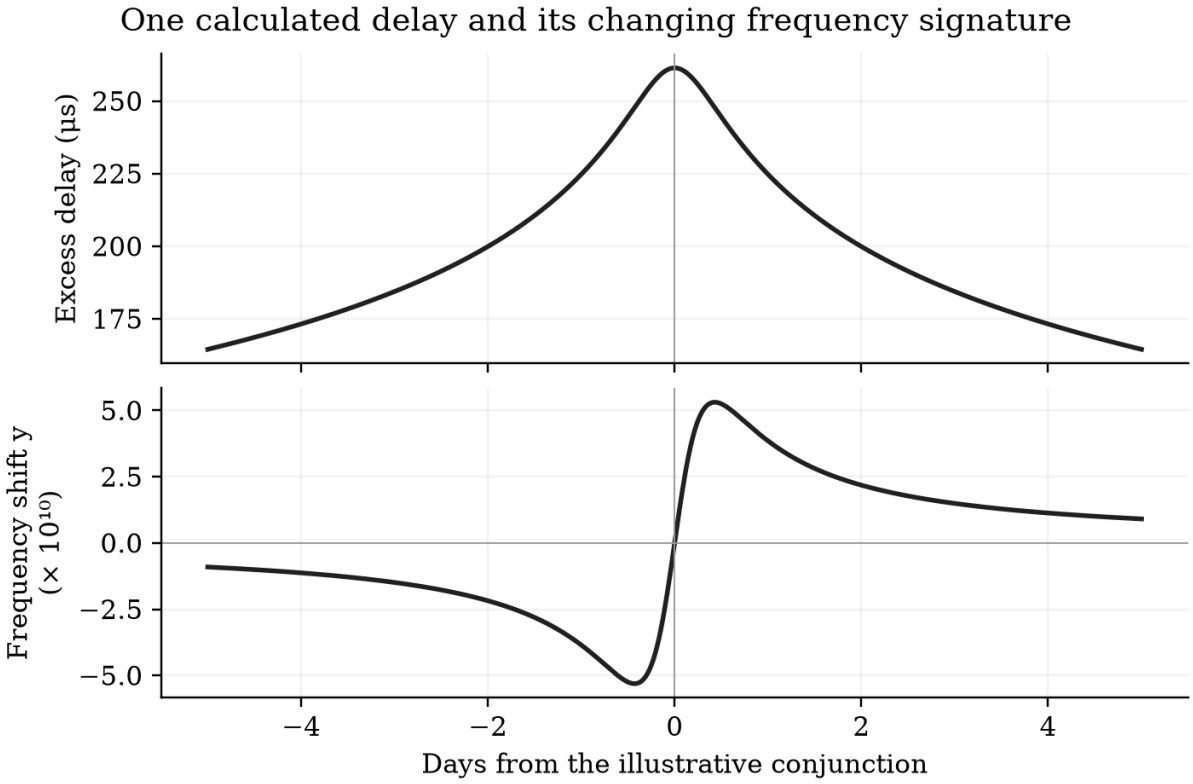


Figure 1. The same calculated propagation delay and its frequency signature. The upper panel shows the excess round-trip time. The lower panel uses the received-frequency sign convention of equation (22). The curves illustrate the relation between delay and its derivative; they are not Cassini measurements.

The empirical comparison uses the parameter actually reported by Bertotti, Iess and Tortora [2]. Their radio-link experiment measured a frequency shift near solar conjunction and obtained

$$\gamma_{\mathrm{obs}} - 1 = (2.1 \pm 2.3) \times 10^{-5} . \quad (30)$$

Here γ is the parametrized post-Newtonian light-propagation coefficient, not a Lorentz factor. At first order, the conventional optical index is $n = 1 + (1 + \gamma)\mu_Q/(c^2\rho)$. Equation (17) fixes

$$\gamma_{\mathrm{SET}} = 1, \ \frac{\gamma_{\mathrm{SET}} - \gamma_{\mathrm{obs}}}{\sigma_\gamma} = -0.913 . \quad (31)$$

Equivalently, the reported delay-amplitude multiplier relative to the retained prediction is $(1 + \gamma_{\mathrm{obs}})/2 = 1.0000105 \pm 0.0000115$. These are two readings of the same measurement, not independent tests. The SET value is within one quoted standard deviation of the published estimate.

The comparison uses the published gamma estimate, rather than refitting Cassini tracking data. The experiment's analysis included multifrequency plasma calibration and spacecraft and station modeling [2, 4]. Our illustrative first-order curves are not a tracking model at that precision. The retained spherical response has the same gamma as GR, so this observation is compatible with that SET branch without distinguishing the two interpretations.

8 Why the joint calculation matters

An orbit and a light path examine different combinations of the response. This mattered in our earlier research: an alternative clock profile could be paired with a radial response that restored the leading perihelion advance, while changing the predicted light bending [5]. The distinction can be written compactly.

For this comparison only, let a general static spherical interval have $g_{TT}/c^2 = A$ and radial coefficient B. With $u = \mu_Q/(c^2 r)$, write

$$A = 1 - 2u + a_2 u^2 + O(u^3), \quad B = 1 + \lambda u + O(u^2). \quad (32)$$

The leading orbit advance and distant-source bending within this comparison class are

$$\Delta\varpi = \frac{\pi\mu_Q}{pc^2}(4 + \lambda - a_2), \quad \delta\theta = \frac{\mu_Q}{bc^2}(2 + \lambda). \quad (33)$$

Here p is the orbit's Kepler-order semilatus rectum, not momentum. The retained response has $a_2 = 0$ and $\lambda = 2$, giving coefficients 6π and 4. The conditional alternative in [5] had $a_2 = 3/2$. Choosing $\lambda = 7/2$ repaired its perihelion coefficient to 6π, but changed the bending coefficient to $11/2$, which is 37.5 percent larger.

The same field must therefore survive more than one kind of measurement. In the construction used here, the source law fixes the source coefficient, the retained exterior fixes the clock and spatial response, and the probe coupling supplies motion and optical paths. The clocks, delay and deflection calculations then remain connected without readjustment.

This is the physical picture SET is presenting: a clock does not stand outside the field and watch it act. Its event progression participates in the same capacity that governs propagation. Local measurements preserve c, while comparisons across the field reveal the burden through clock ratios, accumulated signal delay and changes of direction. Within the spherical response retained here, those consequences are quantitatively compatible with the Cassini light-propagation constraint.

Calculation checks

The accompanying calculation performs 13 symbolic identity checks, including the source conversion, probe energy relation, radial travel-time integral, isotropic coordinate transformation, optical-index expansion and null orbit equation. Independent numerical quadrature reproduces the three finite-endpoint delays in Table 1. Finite differences of those delays reproduce their frequency derivatives. Exact null-ray quadrature approaches the leading bending coefficient as the impact parameter increases.

These checks establish internal calculation consistency. Table 1 and Figure 1 are computed examples; equation (30) is the empirical input. The source strength and optical response are not fitted to that input. Higher-order light-time terms, moving endpoints, planetary fields and plasma corrections would be required for a new analysis of the raw tracking observations.

References

[1] Erik Echeverria. SET perihelion derivation 17 (2026). Retained spherical source, work-current correspondence and probe coupling. See also SET current and motion and SET invariant derivation library, entries A08, A15, A16 and B16 (2026).

[2] B. Bertotti, L. Iess and P. Tortora. A test of general relativity using radio links with the Cassini spacecraft. Nature 425, 374–376 (2003). https://doi.org/10.1038/nature01997

[3] E. E. Mamajek et al. IAU 2015 Resolution B3 on Recommended Nominal Conversion Constants for Selected Solar and Planetary Properties (2015). https://arxiv.org/abs/1510.07674

[4] B. Bertotti. A test of general relativity with radio links with the Cassini spacecraft. Author presentation of the experiment, measurement and calibration. https://www.oca.eu/Mignard/Grex/Presentations_pdf/Grex04_B_Bertotti.pdf

[5] Erik Echeverria. SET capacity energy and propagation research (2026), Appendix F. Conditional comparison of the radial response required by perihelion and light bending.